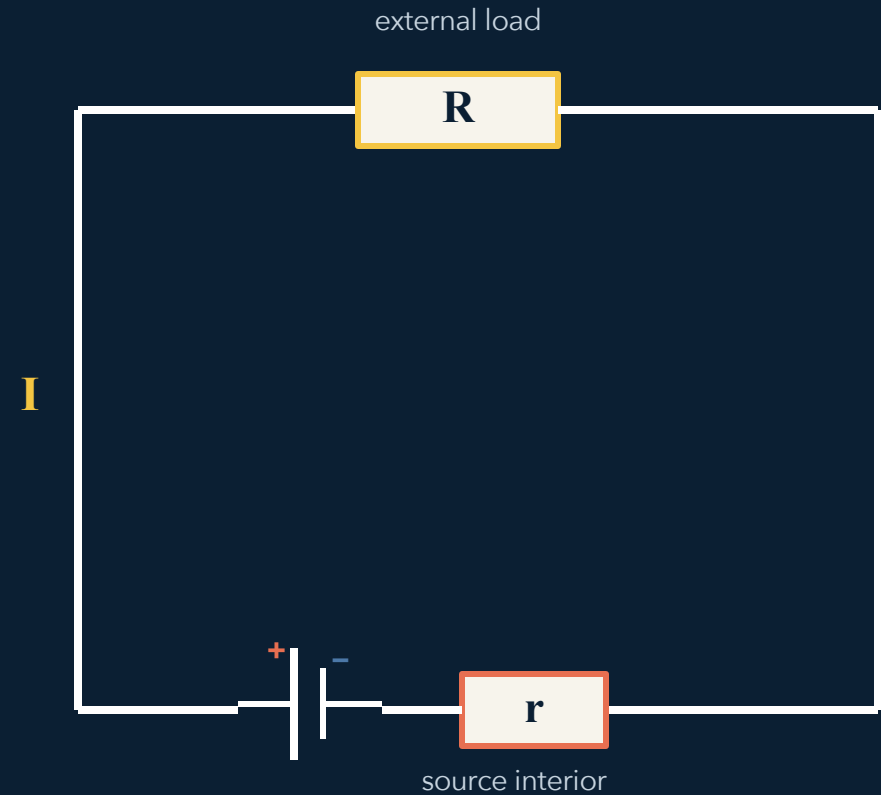


Ohm's Law for Closed Circuits

How a real source shares its voltage between the load and itself



$$\mathcal{E} = I(R + r)$$

Battery heating reveals resistance inside the source.

A real battery is not an ideal voltage source.



When current flows, the battery warms because charges also lose energy inside the source.

$$\text{Internal heat} = I^2 r t$$

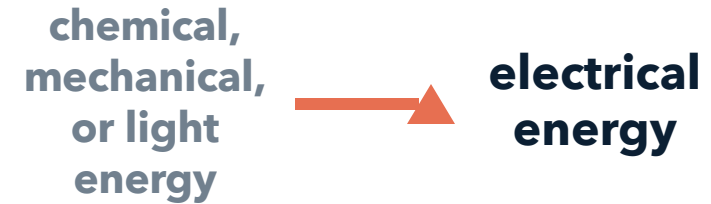
- r is the internal resistance of the source.
- The load resistance R is outside the source.

EMF measures how strongly a source can supply energy.

It is a property of the source, not a measure of stored charge.

$$E = W / q$$

1 volt = 1 joule per coulomb



Non-electrostatic forces move charge from lower to higher electric potential inside the source.

EMF supplies energy; terminal voltage delivers what remains.

Both use volts, but they describe different energy transfers.

INSIDE THE SOURCE

EMF $\mathcal{E}$

energy added per coulomb by non-electrostatic forces

ACROSS THE TERMINALS

Terminal voltage U

energy delivered per coulomb to the external circuit

$$U = \mathcal{E} - Ir$$

Each coulomb receives E , then spends it across R and r .

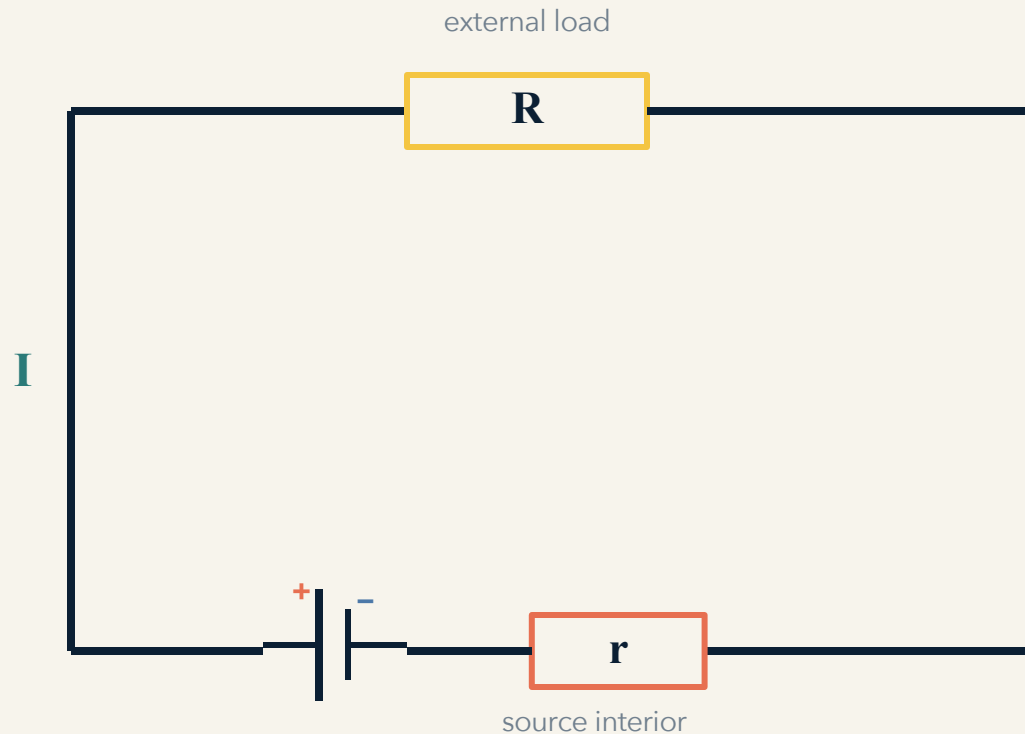
The full-circuit equation follows directly from energy conservation.



$$E q = I R q + I r q \rightarrow E = I(R + r)$$

Current is limited by both the load and the source.

Treat a real source as an ideal EMF E in series with internal resistance r .



$$I = E / (R + r)$$

- Increasing R reduces the current.
- Increasing r also reduces the current.
- If $R \rightarrow 0$, the short-circuit current approaches E/r .

The terminal voltage drops when the source supplies more current.

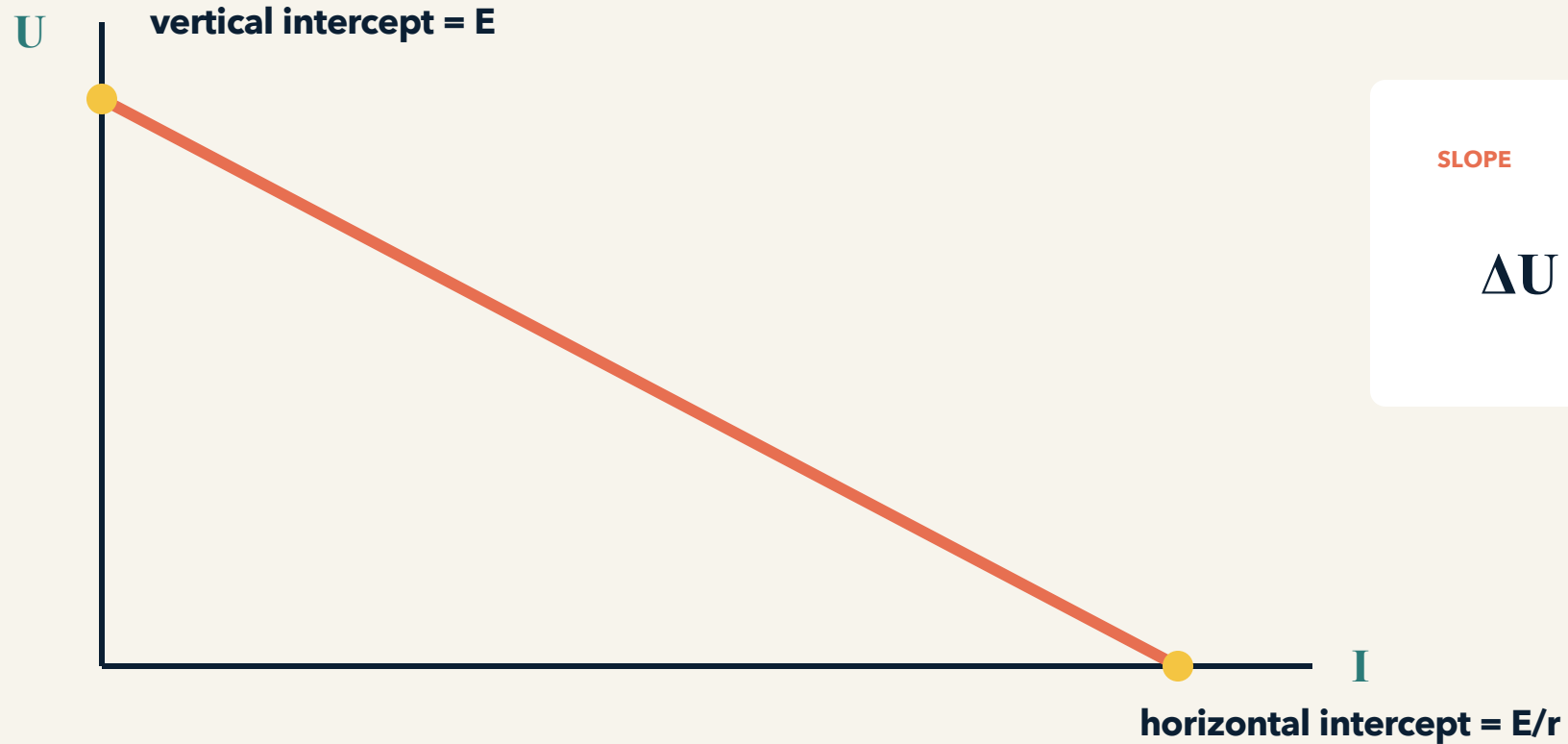
Part of the EMF is lost across the internal resistance.



Open circuit: $I = 0$, so $U = E$. Heavy load: I is large, so U is smaller.

A U-I graph reveals both EMF and internal resistance.

For a source, $U = E - Ir$ is a straight line.



Internal resistance converts part of the source power into heat.

Efficiency improves when the load resistance is large compared with r .

source power

$$P_{\text{source}} = E I = I^2 R + I^2 r$$

USEFUL OUTPUT

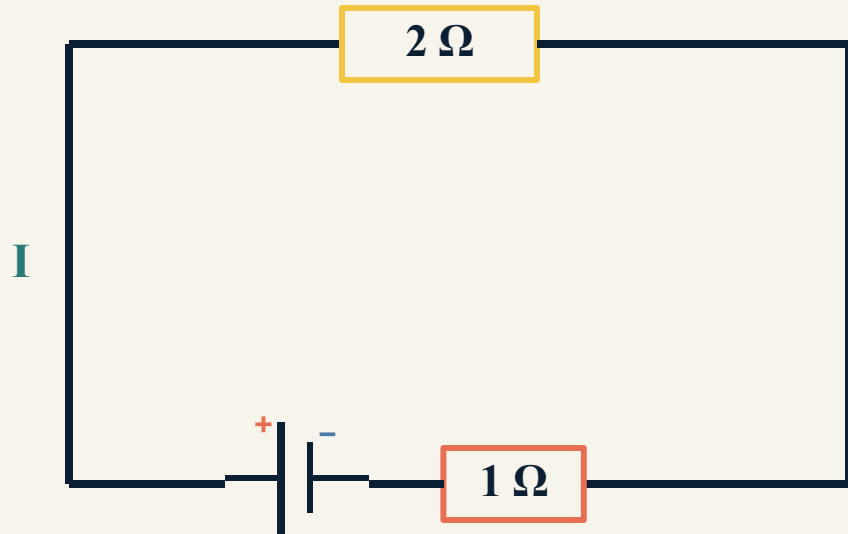
$$P_{\text{load}} = I^2 R$$

EFFICIENCY

$$\eta = R / (R + r)$$

One model answers current, voltage, and heating questions.

Given $E = 3.0 \text{ V}$, $R = 2.0 \Omega$, and $r = 1.0 \Omega$.



1

Current

$$I = E/(R+r) = 3.0/3.0 = 1.0 \text{ A}$$

2

Terminal voltage

$$U = IR = 1.0 \times 2.0 = 2.0 \text{ V}$$

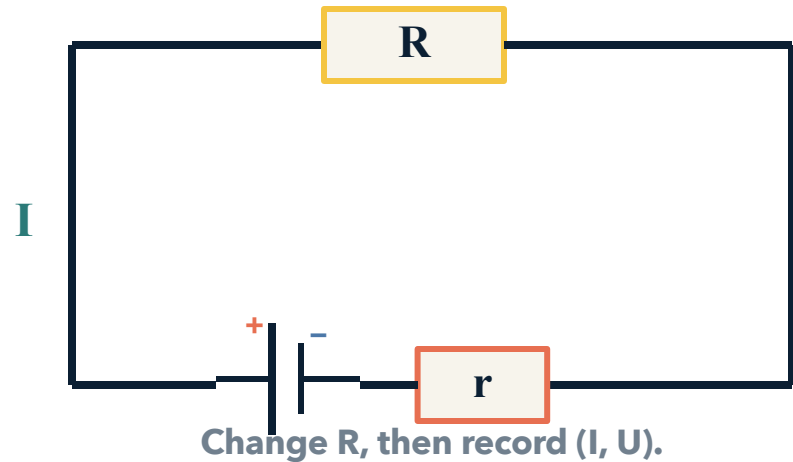
3

Internal heating

$$P_{\text{internal}} = I^2 r = 1.0 \text{ W}$$

Two operating points are enough to determine the source.

Measure terminal voltage and current for two different loads.



Measurement 1

$$U_1 = E - I_1 r$$

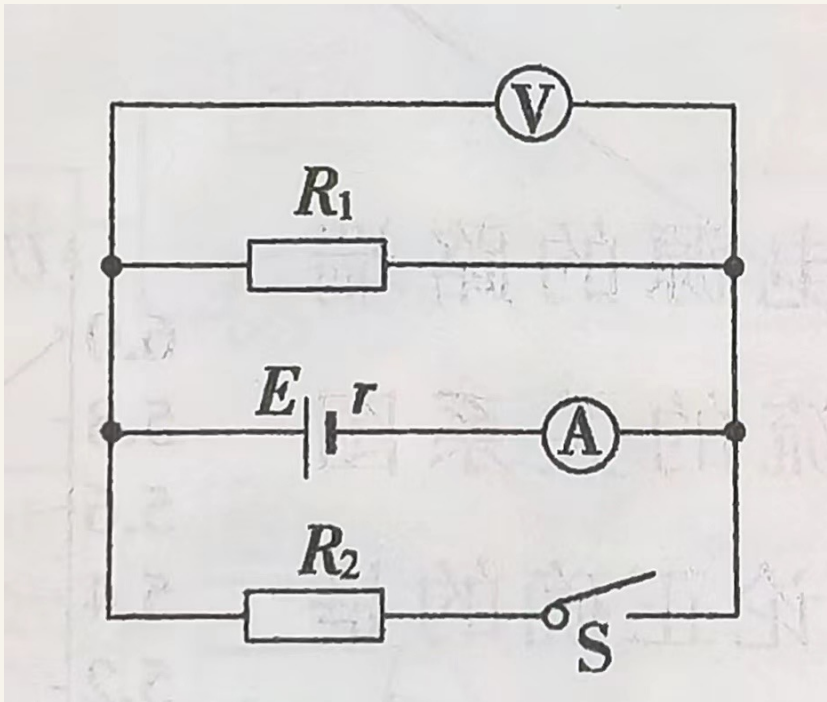
Measurement 2

$$U_2 = E - I_2 r$$

$$r = (U_1 - U_2) / (I_2 - I_1)$$

Use open-circuit voltage and loaded current to identify the source.

An ideal voltmeter and ammeter are connected as shown.



Data

- Switch closed: $U = 1.6 \text{ V}$, $I = 0.4 \text{ A}$
- Switch open: $U = 2.0 \text{ V}$

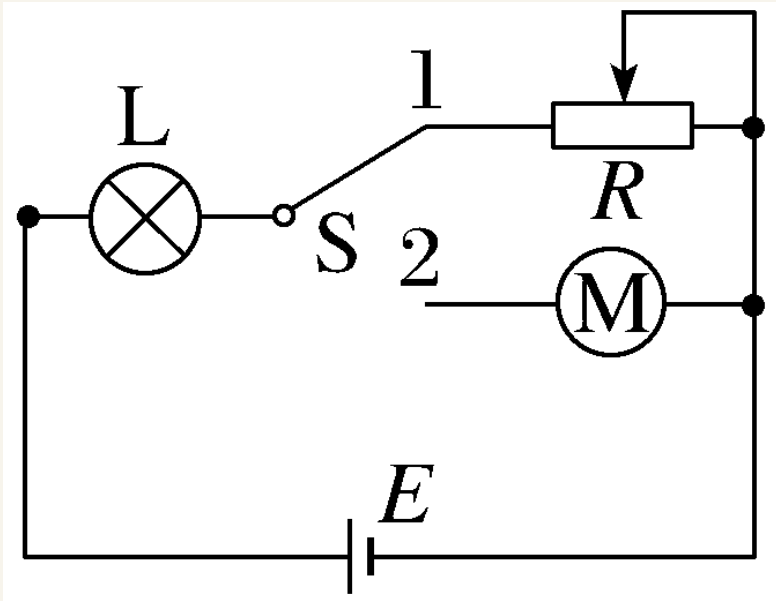
Solution

$$E = 2.0 \text{ V}$$

$$r = (E - U)/I = (2.0 - 1.6)/0.4 = 1.0 \Omega$$

A motor receives less voltage when internal loss is significant.

The original lamp-and-motor circuit becomes an energy-accounting problem.



Reasoning path

- Use the lamp rating to find the circuit current.
- Apply $E = I(R_{\text{total}} + r)$ in each switch position.
- Motor input: $P_{\text{in}} = U_{\text{motor}} I$.
- Motor output = input – coil heating.

Which statement about EMF is correct?

Choose one answer, then justify it using energy per charge.

A. EMF is simply the terminal voltage.

B. A larger EMF means the source must store more energy.

C. A larger EMF means the source must do work faster.

D. EMF equals the energy supplied by the source per unit charge.

Answer: D

A real source supplies energy and consumes some of it.

Model the source, conserve energy, then interpret the result.

MODEL

ideal EMF $\mathcal{E}$
+ internal resistance r

CURRENT

$$I = \mathcal{E} / (R + r)$$

TERMINAL VOLTAGE

$$U = \mathcal{E} - Ir = IR$$

EFFICIENCY

$$\eta = R / (R + r)$$

The central question is always: where does each joule per coulomb go?